ARK SRIVASTAVA

AI / ML Engineer · B.Tech Computer Science & Engineering · Graduating 2026

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PROFESSIONAL SUMMARY

Motivated B.Tech Computer Science student (Galgotias University, 2026) with hands-on experience designing and deploying end-to-end machine learning pipelines, evaluating supervised learning models, and integrating large language model (LLM) APIs into production applications. Proficient in Python, Scikit-learn, XGBoost, Pandas, NumPy, and Flask-based ML deployment. Built a clinical heart disease prediction system benchmarking SVM, Random Forest, Logistic Regression, and XGBoost using ROC-AUC, F1-Score, Precision, and Recall — and deployed it as a Flask web application with real-time risk classification. Developed an AI-powered expense tracker integrating the OpenAI API for natural-language financial insights. Passionate about supervised learning, feature engineering, model optimisation, and applying AI/ML to solve real-world problems.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, Supervised Learning, Classification, Regression, Model Selection, Hyperparameter Tuning, Cross-Validation, Ensemble Methods, Baseline Modelling

Model Evaluation Metrics: ROC-AUC, F1-Score, Precision, Recall, Accuracy, Confusion Matrix, Classification Report, Probability Calibration

Data Science & EDA: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Feature Engineering, Data Preprocessing, Data Cleaning, Dataset Merging, Statistical Analysis

AI & LLM Integration: OpenAI API, Prompt Engineering, LLM-powered features, Generative AI, REST API Integration, JSON Data Pipelines, Third-Party AI Tool Embedding

ML Deployment & Tools: Flask (ML API Deployment), Joblib (Model Serialisation), Jupyter Notebook, Postman, Git, GitHub, VS Code, Netlify

Programming Languages: Python (primary · ML/data science), JavaScript (ES6+), Java

Databases: MySQL, MongoDB — data ingestion and preprocessing for ML workflows

AI / ML PROJECTS

Heart Disease Prediction System · *Python · Scikit-learn · XGBoost · Pandas · NumPy · Matplotlib · Flask · Joblib*

- Engineered a complete end-to-end machine learning pipeline: merged three clinical datasets (UCI Cleveland, Kaggle Heart Disease, supplementary data) into a unified corpus, performed data cleaning, handled missing values, and applied feature preprocessing using Pandas and NumPy.
- Trained and benchmarked four supervised classification models — Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost — evaluating each on Accuracy, Precision, Recall, F1-Score, and ROC-AUC; automatically selected and serialised the best-performing model using Joblib.
- Implemented feature importance extraction using both coefficient-based (logistic/SVM) and tree-based (Random Forest/XGBoost) methods to identify the top clinical predictors driving model decisions.
- Deployed the trained model as a Flask REST API web application providing real-time inference, four-tier risk banding (Low / Moderate / High / Critical), explainable clinical findings, patient-specific recommendations, and comparative ROC curve visualisations — demonstrating production-grade ML deployment.
- Applied a rule-based baseline model as a performance lower bound, establishing a rigorous evaluation framework standard in industry ML workflows.

Expenzo — AI-Powered Expense Tracker · *Python · OpenAI API · Rule-Based AI · Data Visualisation · JavaScript*

- Designed and built an AI-integrated personal finance application using rule-based classification logic and smart categorisation to automatically label expenses by type and pattern — a practical example of lightweight ML-style inference without a heavy model overhead.
- Integrated the OpenAI API to generate natural-language financial summaries and intelligent spending suggestions, applying prompt engineering and LLM API usage in a real production application.
- Built interactive data visualisation dashboards (charts, trend lines) to surface AI-generated insights to end users, completing the full feedback loop from raw data to actionable output.

Sudoku Solver — Constraint Satisfaction & Search AI · *JavaScript · Backtracking · CSP · Algorithm Visualisation*

- Implemented a constraint-satisfaction problem (CSP) solver using recursive backtracking search with forward-checking, solving Sudoku puzzles autonomously — a foundational AI search algorithm used in planning, scheduling, and optimisation systems.
- Visualised each step of the AI decision tree in real time, translating algorithmic reasoning into an interpretable user interface and demonstrating skills in algorithm explainability.

Smart Route Planner — Pathfinding & Optimisation · *JavaScript · Graph Search · Pathfinding Algorithms · Optimisation*

- Designed and implemented a custom graph-based pathfinding and route optimisation algorithm, applying AI-adjacent search logic (similar to A* / Dijkstra) to solve real-world spatial problems.
- Engineered dynamic route recalculation in response to changing user inputs, simulating the adaptive decision-making loops found in reinforcement learning and real-time optimisation systems.

EDUCATION

Galgotias University, Greater Noida, Uttar Pradesh *2022 – 2026 (Expected)*

Bachelor of Technology — Computer Science and Engineering

- Relevant Coursework: Artificial Intelligence, Machine Learning, Python Programming, Data Structures & Algorithms, Database Management Systems, Computer Networks, Statistics

Ramjas School, New Delhi *2019 – 2021*

Class XI – XII · Science Stream (Physics, Chemistry, Mathematics)

ADDITIONAL INFORMATION

GitHub: github.com/ArkSrivastava

Portfolio: arksrivastava.netlify.app

Areas of Interest: Machine Learning · Deep Learning · Generative AI · LLM Integration · NLP · AI Automation · ML Model Deployment · Data Science

Availability: Immediate joiner

Languages: English (Fluent — oral and written) · Hindi (Native)